

CODE SPORT



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In this month's column, we continue our discussion on natural language understanding in the context of dialogue systems.

In last month's column, we had discussed the problem of intent identification in conversational systems. Typically, a dialogue designer who is familiar with the domain will be tasked with listing down the intents that the conversational system needs to cover. However, if there are a large number of intents, some of the infrequent ones can get missed out during the dialogue design process. One way of overcoming this issue is to perform automatic intent mining on the past history of human-to-human conversations in the domain, for which the conversational chatbot system is being designed. If intents can be mined or recognised automatically from the past history of human-to-human conversations, it would very much simplify the work of the dialogue designer. More importantly, it will completely prevent the problem of 'missing intents'.

There is a widespread belief among chatbot designers that by simply adding more example utterances as input to a Natural Language Understanding system, intent recognition accuracy can be improved. However, this is incorrect. Let's assume that certain dialogue designers never considered a couple of intents while designing a particular conversational system. Later, no matter how many example utterances you add for other intents, the system can never recognise these missing intents (this is also the case with a missing category label in a multi-class categorical classification problem). Automatic intent mining can help mine any possible missing intents that dialogue designers may have forgotten. Now, let us look at how automatic intent mining can be performed.

In last month's column, we had posed the following problem to our readers: you are given a set of questions, and there are two related tasks — identifying similar questions and identifying duplicate questions. Which of these two tasks is related to the task of automatic intent mining? The answer is, both. The similar questions task is about identifying questions that are semantically related or, in other words, those that are related to the same topic. The duplicate questions task is about identifying questions

that are exact duplicates. While these two tasks are closely related to the task of intent mining, they are not the same. Let us look at an example.

Q1: How can I renew my deposit account?

Q2: How can I open a new deposit account?

Questions Q1 and Q2 are similar, but they are not duplicate questions. Q1 has the intent of renewing a deposit account whereas Q2 has the intent of opening a new deposit account. These two questions have two different intents.

Q3: How can I open a new deposit account?

Q4: Can you help open a new deposit account?

Questions Q3 and Q4 are not just semantically related — they are exact duplicate questions. They both have the same intent of opening a new deposit account.

Q5: How many flights are there in the morning from Florida to New York?

Q6: How many flights are there in the morning from New York to Florida?

Q5 and Q6 are not duplicate questions since the answers to these two questions are completely different.

Q7: Can you book me a flight in the morning from Florida to New York?

Q8: Can you book me a flight in the morning from New York to Florida?

Now Q6 and Q7 are not duplicate questions since Q7 and Q8 are asking different tasks to be completed. Note that both Q7 and Q8 have the same intent, namely, 'booking a flight' but the entity/slot values associated with the intent are different. For Q7, the origin location is Florida and the destination is New York, whereas for Q8, the origin location is New York while the destination is Florida. Since the entity values are different, these two are not duplicate questions. For the latter, it is not enough that just the intents of the two questions match, but also that the entity values for all the slots should match.

Now that we understand how the task of mining conversational intents is related to similar question retrieval and duplicate question detection, let us look